

Rolina Qu

Aldie, VA | +1 571 544 1227 | rolina.v.qu@gmail.com | <https://github.com/rolinaqu>

OBJECTIVE

Computer Science and Engineering graduate with experience in software development, machine learning research, cloud computing, and high-performance computing seeking a full-time Software Engineer or Computer Scientist role.

EDUCATION

The Ohio State University College of Engineering, Columbus, Ohio 3.6 GPA
B.S. in Computer Science and Engineering, minor in Studio Art. Graduated December 2025.
Honors: Alpha Lambda Delta Honor Society, Dean's List (6 semesters)
Earned the Machine Learning Certificate from Cornell University's eCornell Professional Certificate Program
U.S. Citizen, Eligible to Obtain a Security Clearance

QUALIFICATIONS

- Strong foundation in object-oriented programming, computer systems, data structures and algorithms, machine learning, data science, and software engineering best practices.
- Languages: Java (6+ years), Python (6 years), experience with Linux, Windows, C#, C, HTML, CSS, SQL, noSQL, Bash
- Frameworks and Libraries: JUnit, NumPy, PyTorch, TensorFlow, Pandas, Keras, Matplotlib, Jupyter, SciPy
- Concepts: Agile, object-oriented design, full-stack development, cloud computing, high-performance computing
- Fluent in English and Chinese, intermediate level in Spanish.

ACADEMIC PROJECTS

Neural Collapse Research, May-Aug 2025

- github.com/rolinaqu/distillation-collapse
- Conducted independent research on dataset distillation techniques by leveraging neural collapse theory to improve representational efficiency and training performance of deep neural networks
- Designed and implemented experimental pipelines in Python using deep learning architectures such as ResNet-18 and benchmark datasets like CIFAR-10 to train and evaluate models on distilled datasets
- Leveraged the Ohio Supercomputing Center's high-performance computing cluster to run large-scale experiments, manage distributed jobs, and analyze computational performance

Pathfinder Full-Stack Route Planning Application, Aug-Dec 2025

- github.com/rolinaqu/pathfinder-map-project
- Developed and deployed a full-stack application on AWS in a 5-person team using the NextJS web interface and API
- Managed complex codebase using Github and maintained 4,000+ lines of code over several months
- Designed a recommendation system using Claude AI to suggest and generate relevant stops along routes
- Implemented user authentication with Google Sign-In (OAuth 2.0) and managed persistent data storage using Supabase
- Participated in Agile development, organized team meetings, and documented project features using Github
- Successfully presented final application to real-world users and demonstrated real-time use cases

Speech Emotion Recognition Project, Aug-Dec 2025

- github.com/rolinaqu/The_Sentiment_Squad
- Developed a speech emotion classification system in Jupyter Notebooks using deep learning models trained on multiple benchmark datasets, including CREMA-D, RAVDESS, SAVEE, and TESS
- Implemented audio preprocessing, data augmentation techniques, and feature extraction pipelines using Librosa and NumPy
- Designed and trained multiple neural network architectures, including CNNs, RNNs, and LSTMs
- Evaluated model performance using Scikit-learn metrics and visualized results with Matplotlib and Seaborn
- Achieved 86% classification accuracy using a CRNN with ReLU and softmax activations. Determined that Mel-frequency filter banks and MFCCs were the most informative features

WORK EXPERIENCE

Undergraduate Research Apprenticeship Program, May-Aug 2025:

- Worked with Dr. Andrew Perrault and Dr. Zhihui Zhu to perform novel research on machine learning methods using the Ohio Supercomputer Center (OSC)'s computing clusters
- Developed novel constraints for data distillation methods and wrote bash scripts to test and run code
- Presented research findings and abstract at the university's Spring Undergraduate and Graduate Research Symposium